

Simra Ali

Computer Engineering · Robotics, Autonomous Systems & Embedded Design

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#1 in MA · #11 in the world FTC 3565, out of 7,800+ teams (2026)	3× 1st place Hackathons & CAD competitions, 2024–25	GPA 4.18 · SAT 1560 790 Math · 770 EBRW	MIT BWSI · Harvard SSP Autonomous systems coursework
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EDUCATION

Westford Academy · Westford, MA Class of 2027

GPA 4.18 · SAT 1560 (790 Math, 770 EBRW) · Academic High Honors, 2023–present

Coursework: AP Calculus BC, AP Physics C, AP Chemistry, AP Computer Science A, AP Biology, AP U.S. History, AP Psychology, AP Microeconomics

MIT Beaver Works Summer Institute · Lincoln Laboratory Feb 2024 – Aug 2026

Autonomous Air Vehicle Racing	on-site · summer	Basics of ASICs	instructor led · fall
Microelectronics & Hardware Development	online	Python Core	
Version Control: Git/GitHub & UNIX			

Harvard Summer School · Cambridge, MA Summer 2025

Robotics, Autonomous Vehicles, Drones, and Artificial Intelligence — **Grade: A**

ROBOTICS LEADERSHIP

FIRST Robotics Competition · Stormgears 5422 (45 members) 2023 – Present

Co-Captain · CAD / Mechanical Design Lead · Drive Team Coach · Former Electrical Lead

- **Taught myself SolidWorks** after the team's experienced designer graduated, then helped lead a custom-robot effort that earned the **FIRST Quality Award** for design intent, robustness, and fabrication quality.
- Designed competition mechanisms in SolidWorks — chassis, intake, hopper, shooter, electrical board, belly pan, and component holders — specified for serviceability between matches.
- As **Electrical Lead**, ran summer and fall training, managed inventory, scheduling and task tracking (Slack, Trello), and trained the student who succeeded me in the role.
- Drive Team: Coach at Rebuilt 2026 and Technician at 2025 New England District Championship; Driver at the 2024 Clipper Clash.
- Presented to sponsors including **IBM, Triton Systems and SolidWorks**.

FIRST Tech Challenge · Ghost Robotics 3565 (10 members) 2023 – Present

Vision Software · Onshape CAD

- **2026 Massachusetts State Champion and Winning Alliance Captain; Ross Division Winner at the FIRST World Championship** — team ranked **#1 in Massachusetts and #11 in the world out of 7,800+ teams**.
- Work deliberately across vision software, CAD, strategy, media and outreach rather than limiting myself to one formal title.

ENGINEERING PROGRAMS & RESEARCH

Autonomous Air Vehicle Racing · MIT Beaver Works Summer Institute, AeroAstro Summer 2026

Four-week onsite program

- Designed, programmed, repaired, tested and raced autonomous drones; studied computer vision, state estimation, control systems, navigation and flight dynamics.
- Explored **socially aware path planning** for autonomous drones and connected with an MIT PhD researcher working in the field.
- **Honorable Mention**, BWSI “You Belong in STEM” Essay Contest.

Robotics, Autonomous Vehicles, Drones & AI · Harvard Summer School Summer 2025

Grade: A

- Studied sensors, motion planning, SLAM, reinforcement learning, autonomous systems and computer vision.
- Co-led the “Solve It, Defend It!” innovation club and developed **Flood Watch**, a concept for predicting flood risk from drone and autonomous-vehicle data.

Women in Technology Program · BAE Systems

Jan – May 2026

- Hands-on labs, workshops, tours and presentations spanning electrical, microwave/RF, software, mechanical, signal processing, optical, manufacturing and failure-analysis engineering.

Summer Immersion & Pathways Programs · Girls Who Code

Jun 2024 – Jul 2025

Cybersecurity, cryptography, data science, machine learning, game design and web development.

PROJECTS & COMPETITIONS

The Bat · GirlHacks at Microsoft NERD — 1st Place

Jun 2025

- Assistive navigation device for blind and low-vision users: ultrasonic sensors detect obstacles and a buzzer returns auditory feedback. Designed, built and demonstrated within the hackathon.

HireAble · NetScout Tech for Good — 1st Place

Mar 2025

- Web platform connecting people with disabilities to inclusive employers, integrating database and web technologies to make hiring more accessible.

VTXCADs CADathon · Phillips Exeter Academy Robotics — 1st Place

Dec 2024

Firefly · Girls Who Code game design project

Jul – Aug 2025

- Interactive game built with the p5play / p5.js JavaScript libraries, using **TensorFlow.js hand-gesture detection** as the input method.

SolidWorks Demonstration Robot · built for team sponsor

May – Sep 2024

- FRC-style robot with autonomous vision to locate tennis balls, an intake to collect them, and a shooter to launch them at a target.

OUTREACH, MENTORSHIP & LEADERSHIP

Outreach Coordinator · Middlesex County Robotics Club, Boxboro, MA

Aug 2023 – Present

- Organize and lead Girl Scout STEM Fest, FIRST LEGO League trainings and scrimmages, the Apple Blossom Parade, and elementary and middle school STEM events.
- Mentored an all-girls FTC team** and led the upgrade of the SaturnReach outreach robot; emphasis on getting girls into technical roles and keeping them there.

Co-Founder & Co-Lead · Women in Computer Science, Westford Academy

Sep 2024 – Present

- Teach Python, support member projects, run hackathon preparation, and share STEM opportunities that are easy to miss if nobody tells you they exist.

Manager · Westford Academy Broadcasting Club

Sep 2024 – Present

- Coordinate club operations while working the shows as anchor and production crew: teleprompter, sound board, video switching, scripts and school-wide broadcasts.

Cashier · Demoulas (Market Basket), Westford, MA

Jul 2023 – Dec 2024

HONORS & AWARDS

FIRST Leadership Award (Dean's List) Semi-Finalist	2026	Academic High Honors, Westford Academy	2023–26
BWSI "You Belong in STEM" Essay — Honorable Mention	2026	AP Scholar Award	2026
GirlHacks at Microsoft NERD — 1st Place	2025	FRC Quality, Team Spirit & Gracious Professionalism	2025–26
NetScout Tech for Good — 1st Place	2025	FTC Control, Inspire, Innovate & Connect	2025–26
VTXCADs CADathon — 1st Place	2024	Diversity, Equity & Inclusion Award, Westford	2023
Toastmasters Youth Leadership Award & Medal	2021	Honored Citizen Award, Westford Public Schools	2023

TECHNICAL SKILLS

CAD / Mechanical SolidWorks, Onshape, 3D printing, Autodesk Tinkercad

Hardware Arduino, wiring, soldering, crimping, electrical-board layout, sensors, systematic troubleshooting

Software Python, Java, JavaScript, HTML/CSS, SQL, Git/GitHub, WPILib, TensorFlow, NumPy, p5.js

Robotics / AI Computer vision, SLAM, motion planning, reinforcement learning, autonomous systems, state estimation